

(Everything You Never Knew You Needed To Know About) Rank-Biased Measurement For Web Search

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Abstract

Measurement is a fundamental human activity. It is through measurement that we identify opportunity, and via opportunity that we seek improvement. In information retrieval and web search we measure how good search engines are at ordering answers to user queries, how close two ranked lists are to each other, how good LLMs are at re-ranking sets of candidate documents, and how close generated answer sentences are to the ideal output.

This talk begins by motivating the well-known “top-weighted” measurements that are used when assessing the quality of a ranked list of documents against known annotations (qrels). In particular, Moffat and Zobel [2] suggest use of a parameterized geometric distribution to assign weights to ranks, arguing that a distribution with a bounded infinite sum is preferable to ones with unbounded sums; and also motivating the geometric sequence via a proposed *user browsing model*, the first time this had been done. Their *rank-biased precision* mechanism has the further advantage of allowing the extent of the uncertainty to be quantified, where uncertainty arises whenever the qrels are not comprehensive.

The same user browsing model was next applied to top-weighted comparisons of ranked lists, with *rank-biased overlap* [4] now well-known as a way of computing a similarity score between two sequences that might not be permutations of each other, and might not even be of the same length. There is again a parameter ϕ that reflects the “persistence” of the population of stochastic users assumed to be examining the two rankings.

The second part of the talk then presents recent work unifying those two “rank-biased” approaches as elements in a larger framework that exposes a third rank-biased measurement, *rank-biased recall* [3], a dual of rank-biased precision in the same way that the classic set measurements recall and precision are duals of each other. Applications of this latest rank-biased technique that demonstrate its utility will also be presented and discussed.

Finally, in the third part of the talk, new ways in which the degree of top-weighting bias can be more flexibly controlled will be described (work currently under review [1]). In particular, rank-biased weighting schemes that make use of two-parameter translated inverse power law distributions will be motivated and explored. The virtue of these distributions is that they allow the most important “top” weight to be specified, plus either a target residual at a stipulated depth to be achieved, or, in the case of finite distributions, a

fixed lowest weight at a given maximum depth to be realized. The two-dimensional search process that is required to identify suitable parameters will also be sketched.

In conjunction, the suite of rank-biased techniques allow practitioners and researchers to better define their measurement goals and intentions, and hence more precisely target their experiments.

CCS Concepts

• Information systems → Retrieval effectiveness.

Keywords

Evaluation; measurement; top-weighted; rank-biased; precision; overlap; recall

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Biography

Professor Alistair Moffat is now in his 40th year as a faculty member at the University of Melbourne. Early work included coauthorship of the book “*Managing Gigabytes: Compressing and Indexing Documents and Images*” (1994, 1999), and development of innovative mechanisms for implementing ranked queries via compressed inverted indexes. Most recently Alistair has been focused on IR evaluation, including models for user query formulation and result perusal, and the material on rank-biased measurement that is the subject of this talk. Alistair was inducted into the SIGIR Academy in 2021, and recently became a Fellow of the ACM for his contributions to the implementation and evaluation of search systems.

Acknowledgment

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References

- [1] A. Moffat. Targeted parameter selection for rank-biased measurement. Feb. 2026. Under review.
- [2] A. Moffat and J. Zobel. Rank-biased precision for measurement of retrieval effectiveness. *ACM Trans. Inf. Sys.*, 27(1):2.1–2.27, 2008.
- [3] A. Moffat, J. Mackenzie, A. Mallia, and M. Petri. Rank-biased quality measurement for sets and rankings. In *Proc. SIGIR-AP*, pages 135–144, 2024.
- [4] W. Webber, A. Moffat, and J. Zobel. A similarity measure for indefinite rankings. *ACM Trans. Inf. Sys.*, 28(4):20.1–20.38, 2010.



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